

Operating System Types -- Guided Notes

Unit tp-3.1 | CompTIA Network+ N10-009 Obj. FC0-U71 3.1 | N10-008 FC0-U71 3.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

An operating system (OS) is the software that manages a computer's _____ resources and provides services to application software. Without an OS, application programs would have to individually control every piece of _____ themselves. The OS acts as a _____ between hardware and applications.

Question 2

A _____ OS is designed for multi-user, multi-tasking environments on shared hardware. A _____ OS (like Windows 10/11) is designed for single-user productivity on personal computers. A _____ OS (like Android/iOS) is optimized for battery life and touchscreen interaction on handheld devices.

Question 3

Windows is developed by _____ and holds approximately _____ % of the desktop OS market. Windows uses a _____ (graphical) interface by default. The Windows Registry is a centralized database that stores _____ settings for the OS and applications.

Question 4

macOS is developed by _____ and runs on _____ hardware (primarily Apple Mac computers). It is built on a _____ foundation (derived from BSD Unix). This gives macOS access to a _____ terminal and Unix-compatible command-line tools.

Question 5

Linux is an open-source OS kernel first created by _____ in 1991. A Linux _____ is a complete OS built around the Linux kernel plus additional software. Examples include Ubuntu, _____, and _____. Linux is dominant in _____ and cloud infrastructure.

Question 6

Chrome OS is developed by _____ and is built on the _____ kernel. It is designed primarily to run _____ and Android apps. Chrome OS is popular in _____ because devices are inexpensive and easy to manage centrally via Google Admin console.

Question 7

An embedded OS (like _____ RTOS or VxWorks) runs inside specialized hardware with no GUI. Mobile OSes include _____ (Google, open source) and _____ (Apple, closed). The key difference between mobile and desktop OSes is that mobile OSes are optimized for _____ efficiency and _____ interfaces.

Question 8 -- Real World Application

REAL WORLD: A school district is choosing an OS for 500 new student laptops. Options are Windows 11, Chromebook (Chrome OS), and Ubuntu Linux. The laptops will be used for web browsing, Google Classroom, and basic word processing. Budget is tight. Which OS would you recommend and why? Include at least two reasons.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-3-1-operating-system-types/tp-3-1-operating-system-types-answer-key.pdf
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